

DLNM-LC RCP Simulation

Setup

```
packages <- c(
  "readxl", "forecast", "dlnm", "splines", "MASS", "astsa"
)
invisible(lapply(packages, library, character.only = TRUE))

find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) return(path)
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}

base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()
available_scenarios <- c("rcp26", "rcp85")
default_params <- list(
  scenario = "all",
  scenarios = available_scenarios,
  nsim = 10000,
  simulation_weeks = 1352,
  run_simulation = TRUE
)

rmd_params <- if (exists("params") && !is.null(params)) {
  modifyList(default_params, as.list(params))
} else default_params

if (!is.null(rmd_params$scenario) && !identical(rmd_params$scenario, "all")) {
  scenarios <- rmd_params$scenario
} else {
  scenarios <- rmd_params$scenarios
}

scenarios <- unname(unlist(scenarios))
if (!all(scenarios %in% available_scenarios)) {
  stop(
    "Unknown scenario(s): ",
    paste(setdiff(scenarios, available_scenarios), collapse = ", ")
  )
}

nsim <- rmd_params$nsim
simulation_weeks <- rmd_params$simulation_weeks
```

```
run_simulation <- isTRUE(rmd_params$run_simulation)

invisible(lapply(list.files(file.path(base_dir, "Code/Function"),
                             pattern = "\\R$", full.names = TRUE), source))
```

Inputs

```
historical_inputs <- build_historical_inputs(base_dir)
dat_list <- historical_inputs$dat_list
wave_list <- historical_inputs$wave_list

cb_list <- load_historical_cb_list(
  base_dir,
  lag_days = 21,
  var_df = 2,
  lag_df = 2,
  degree = 3,
  legacy_lag_knots = TRUE
)
```

Fit DLNM-LC

```
DLNM_LC_models <- lapply(rcp_internal_region_names, function(region) {
  DLNM_LC(
    dat = dat_list[[region]],
    cross_matrix = cb_list[[region]],
    wave_data = wave_list[[region]],
    tol = 1e-2,
    max_iter = 20,
    region = region
  )
})
```

```
## convergedconvergedconverged
```

```
names(DLNM_LC_models) <- rcp_internal_region_names
```

```
dlnm_coef <- lapply(DLNM_LC_models, `[`, "dlnm_coef")
```

```
dlnm_vcov <- lapply(DLNM_LC_models, `[`, "dlnm_vcov")
```

SARIMA Inputs For Kappa

```
sarima_inputs <- lapply(scenarios, function(scenario) {
  write_legacy_sarima_inputs(
    model_type = "DLNM_LC",
    scenario = scenario,
    base_dir = base_dir
  )
})
names(sarima_inputs) <- scenarios
sarima_inputs
```

```
## $rcp26
```

```

##      model time_series parameter      estimate
## 1  DLNM_LC      Attiki      ar1  1.30391560
## 2  DLNM_LC      Attiki      ar2 -0.36531150
## 3  DLNM_LC      Attiki      ma1 -0.97330070
## 4  DLNM_LC      Lisbon      ar1  0.52663241
## 5  DLNM_LC      Lisbon      d1  0.00000000
## 6  DLNM_LC      Lisbon      ma1  0.00000000
## 7  DLNM_LC      Lisbon      sar1 0.08157594
## 8  DLNM_LC      Lisbon      D1  0.00000000
## 9  DLNM_LC      Lisbon      sma1 0.00000000
## 10 DLNM_LC      Lisbon      S1 52.00000000
## 11 DLNM_LC      Roma      ar1  1.06836980
## 12 DLNM_LC      Roma      ar2 -0.10362570
## 13 DLNM_LC      Roma      d1  0.00000000
## 14 DLNM_LC      Roma      ma1 -0.84380670
## 15 DLNM_LC      Roma      sar1 0.09644500
## 16 DLNM_LC      Roma      D1  0.00000000
## 17 DLNM_LC      Roma      sma1 0.00000000
## 18 DLNM_LC      Roma      S1 52.00000000
##
## $rcp85
##      model time_series parameter      estimate
## 1  DLNM_LC      Attiki      ar1  1.30391560
## 2  DLNM_LC      Attiki      ar2 -0.36531150
## 3  DLNM_LC      Attiki      ma1 -0.97330070
## 4  DLNM_LC      Lisbon      ar1  0.52663241
## 5  DLNM_LC      Lisbon      d1  0.00000000
## 6  DLNM_LC      Lisbon      ma1  0.00000000
## 7  DLNM_LC      Lisbon      sar1 0.08157594
## 8  DLNM_LC      Lisbon      D1  0.00000000
## 9  DLNM_LC      Lisbon      sma1 0.00000000
## 10 DLNM_LC      Lisbon      S1 52.00000000
## 11 DLNM_LC      Roma      ar1  1.06836980
## 12 DLNM_LC      Roma      ar2 -0.10362570
## 13 DLNM_LC      Roma      d1  0.00000000
## 14 DLNM_LC      Roma      ma1 -0.84380670
## 15 DLNM_LC      Roma      sar1 0.09644500
## 16 DLNM_LC      Roma      D1  0.00000000
## 17 DLNM_LC      Roma      sma1 0.00000000
## 18 DLNM_LC      Roma      S1 52.00000000

```

DLNM Component And Final Simulated Mortality

```

lc_simulation_outputs <- do.call(rbind, lapply(scenarios, function(scenario) {
  run_lc_rcp_scenario(
    scenario = scenario,
    DLNM_LC_models = DLNM_LC_models,
    dlnm_coef = dlnm_coef,
    dlnm_vcov = dlnm_vcov,
    base_dir = base_dir,
    simulation_weeks = simulation_weeks,
    nsim = nsim
  )
})

```

```
)))
```

```
## Running legacy-compatible DLNM-LC RCP simulation for rcp26
```

```
## Running legacy-compatible DLNM-LC RCP simulation for rcp85
```

```
lc_simulation_outputs
```

```
##      scenario          object
```

```
## 1      rcp26 final_LC.sim.rcp26
```

```
## 2      rcp85 final_LC.sim.rcp85
```

```
##
```

```
## 1 /Users/jiachengabelmin/Desktop/LC_DLNM_paper_repro/Results/Simulated_mortality_rates/final_LC_rcp26
```

```
## 2 /Users/jiachengabelmin/Desktop/LC_DLNM_paper_repro/Results/Simulated_mortality_rates/final_LC_rcp85
```